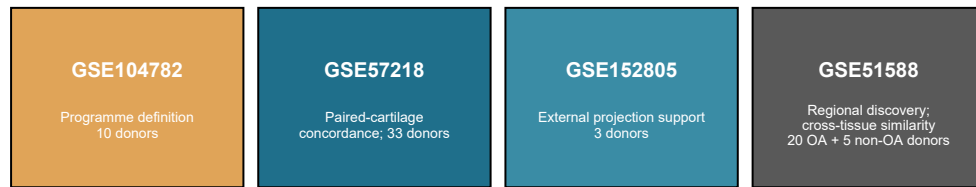


a



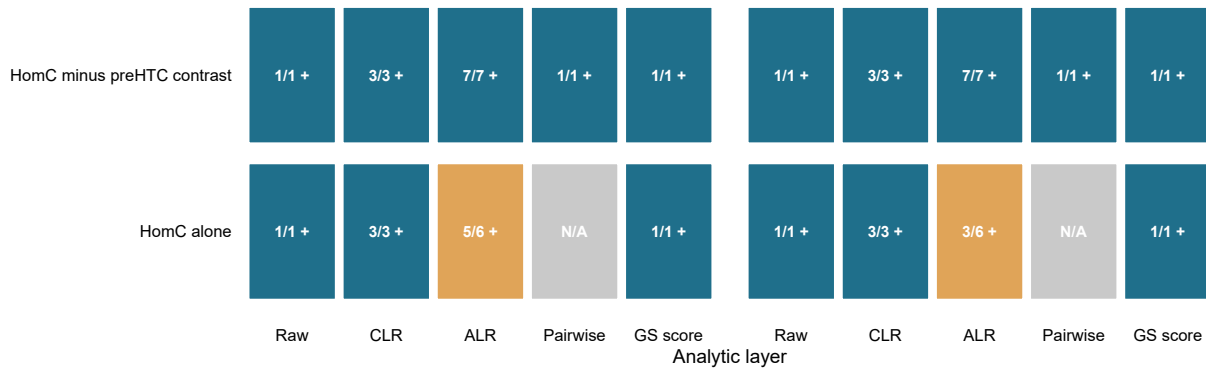
b

The contrast remained positive under every final transformation, zero-handling rule and reference specification in both datasets.

Consistent + Not applicable Reference-dependent

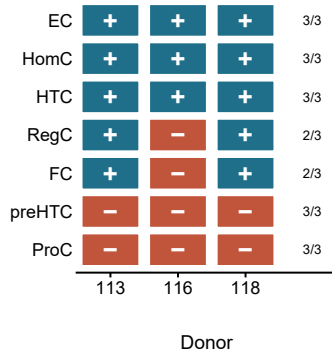
GSE51588 bone (LT - MT)

GSE57218 cartilage (preserved - OA)

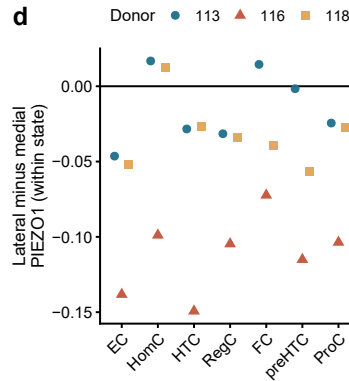


c

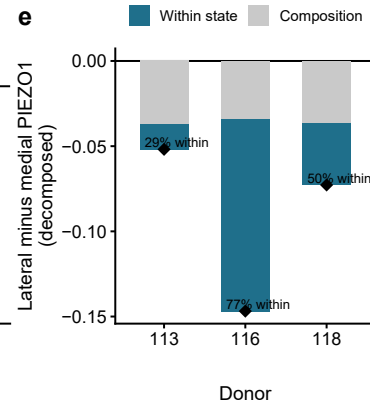
lateral higher medial higher



d



e



f

Direction only; effect magnitudes are not compared across contexts

Negative Positive

GSE104782 | Across-state association

PIEZO1 decreases toward hypertrophic pole; 9/10 donor correlations negative

GSE152805 | Cartilage compartment

Lateral < medial PIEZO1; 3/3 donors

GSE57218 | Cartilage region

Preserved < OA-affected PIEZO1; 33 paired donors

GSE51588 | Subchondral-bone compartment

Lateral > medial PIEZO1; 20 OA paired donors